DIGITAL PAYMENT ADOPTION & RETAIL SECTOR GROWTH: STATE LEVEL UPI ANALYSIS

A Final report for the BDM Capstone Project

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1. Executive Summary

The Unified Payments Interface (UPI) has emerged as a cornerstone of India's digital payments ecosystem since its introduction in 2016, reshaping the way retail transactions are conducted across the country. This study examines whether differences in UPI transaction intensity across Indian states are reflected in variations in Retail Gross State Value Added (GSVA) growth, thereby assessing the role of digital payments in retail sector performance.

The analysis draws on secondary data from authoritative government sources, including state-wise Retail GSVA data published by the Ministry of Statistics and Programme Implementation (MoSPI) for the period 2011–12 to 2023–24, and UPI transaction statistics released by the National Payments Corporation of India (NPCI). Given data availability constraints, national-level UPI trends are examined over time, while state-wise UPI transaction value for the financial year 2023–24 is used to capture relative digital payment intensity across states.

A descriptive and classification-based analytical framework is employed, involving growth rate computation, percentile-based categorization, and an integrated state classification combining retail market size, retail growth momentum, and UPI transaction value share. The findings show that although UPI adoption has expanded rapidly and is concentrated in economically larger states, higher UPI transaction intensity does not consistently correspond to higher Retail GSVA growth. Retail sector expansion was evident prior to the introduction of UPI, indicating that structural economic factors such as income levels, urbanization, and consumption demand remain the primary drivers of growth.

The study further identifies four distinct categories of states, Digitally Mature Markets, Core Growth Engines, Emerging Growth Markets, and Low Adoption & Low Maturity states, each reflecting different combinations of economic scale, growth performance, and digital payment adoption. Overall, the results suggest that UPI functions as a complementary enabler that enhances efficiency and formalisation within the retail ecosystem, rather than as a standalone driver of retail sector growth. The findings underscore the importance of integrating digital payment initiatives with broader economic and retail development strategies to achieve more balanced and sustainable growth across states.

2. Proof of Originality and Data Sources

This project is based entirely on secondary data collected from publicly available and authoritative sources. No primary data collection, surveys, or organisational collaborations were undertaken.

The main data sources include:

- **Ministry of Statistics and Programme Implementation (MoSPI), Government of India**: State-wise Retail GSVA data across multiple financial years.
- **National Payments Corporation of India (NPCI)**: National and state-wise UPI transaction statistics, including monthly and annual transaction volumes and values.

Data Repository References

- MoSPI GSVA data: https://www.mospi.gov.in
- NPCI UPI statistics: https://www.npci.org.in/what-we-do/upi/product-statistics
- NPCI monthly state-wise UPI statistics: https://www.npci.org.in/product/ecosystem-statistics/upi

All datasets were independently cleaned, transformed, and analysed. No figures, analyses, or interpretations were copied from existing studies. Only raw data was sourced from the above repositories, in compliance with the project's originality requirements.

3. Metadata and Descriptive Statistics

This section documents the data architecture, variable definitions, and descriptive characteristics of all datasets used in the study. It establishes transparency regarding data sources, time coverage, variable construction, and initial observable patterns, before any analytical transformations or classification logic are applied. This separation ensures that data understanding precedes methodological inference.

3.1 Data Sources and Dataset Structure

The study is based on two core secondary datasets: Retail Gross State Value Added (GSVA) data and Unified Payments Interface (UPI) transaction data.

Retail GSVA Dataset

- Source: Ministry of Statistics and Programme Implementation (MoSPI), Government of India
- Sector coverage: Retail trade (state-wise)
- Temporal coverage: Financial years 2011–12 to 2023–24
- Data frequency: Annual
- Measurement unit: ₹ lakh / ₹ crore (as published by MoSPI)
- Spatial coverage: Major Indian states; select Union Territories excluded due to incomplete time series

Retail GSVA represents the monetary value added by the retail sector within each state and serves as the primary indicator of retail sector performance. Since GSVA data for certain states was unavailable for 2024–25, all longitudinal analysis was restricted to the 2011–12 to 2023–24 period to maintain comparability across states.

UPI Transaction Dataset

- Source: National Payments Corporation of India (NPCI)
- Components:
    - Monthly national UPI transaction volume (number of transactions)
    - Monthly national UPI transaction value (₹ crore)
    - Annual national UPI aggregates derived from monthly data
    - State-wise UPI transaction value for the financial year 2023–24
- Temporal coverage:
    - National-level data: 2016 onwards
    - State-wise data: FY 2023–24 only

Due to the absence of consistent historical state-wise UPI transaction volume data, state-wise UPI transaction value share was used as an indicator of relative digital payment intensity across states.

3.2 Variable Definition and Construction

Key variables used in the study include:

- **Retail GSVA (₹)**: Annual state-wise retail sector output
- **Absolute GSVA Increase**: Difference between GSVA in 2023–24 and 2011–12
- **GSVA Growth Percentage**: Percentage growth over the study period
- **UPI Transaction Value (₹ crore)**: Total value of UPI transactions
- **UPI Share (%)**: Proportion of national UPI transaction value contributed by each state in FY 2023–24

					NOTES: SINCE SOME STATES DON'T HAVE GSVA DATA OF 2024-25 THESE CALCULATIONS EXCLUDE 2024-25 DATA OF ALL STATES				
		Retail_GSVA (Rs. in lakh)	Column Labels						
		States	**2011-12**	**2023-24**	**Grand Total (All Years)**	**Absolute Increase**	**Growth %**	**Growth Category**	**Size Category**
		Maharashtra	10560912.41	17628320.7	198949667.9	7067408.29	66.9204327 6	Low	High Size

Table 1: GSVA growth and absolute increase variables

Derived variables were computed directly within Excel using arithmetic and logical formulas, ensuring reproducibility of results.

3.3 Data Preparation and Alignment

Prior to descriptive analysis, datasets were aligned to a common financial-year framework. Monthly UPI data was aggregated to annual and financial-year totals where required. States with incomplete GSVA data were excluded from analyses involving growth computation. Unit standardisation was applied to ensure consistent monetary representation across datasets.

Financial YEAR	ANNUAL TRANSACTIONS (in Mn.)	ANNUAL VALUE (Rs. in Cr.)
2016-17	**17.86**	**6952.14**
2017-18	**915.22**	**109831.8**
2018-19	**5391.52**	**876970.72**
2019-20	**12518.61**	**2131730.14**
2020-21	**22330.65**	**4103653.58**
2021-22	**45967.53**	**8417572.48**
2022-23	**83751.14**	**13920676.35**
2023-24	**131129.83**	**19995085.51**
2024-25	**185866.02**	**26056954.65**

Table 2: Annual UPI Data (According to Financial Year)

3.4 Descriptive Statistics: Retail GSVA

This subsection presents the descriptive statistical profile of state-wise Retail GSVA values to summarise the central tendency, dispersion, and distributional characteristics of retail sector output.

Table 3.1 reports the mean, median, standard deviation, minimum, maximum, quartiles, and interquartile range of Retail GSVA across states for the financial year 2023–24. The substantial gap between the mean and median indicates a right-skewed distribution, reflecting the dominance of a few economically large states in national retail output. The wide range between minimum and maximum values highlights significant inter-state variation in retail sector size.

Table 3.2 presents descriptive statistics for the long-term Retail GSVA growth percentages. The dispersion measures indicate considerable heterogeneity in retail sector expansion across states, with some states recording very high growth from smaller bases while others exhibit relatively modest growth. The interquartile range confirms that growth performance varies widely even among the middle group of states.

These descriptive measures establish that both the scale and growth of the retail sector differ substantially across Indian states, justifying the need for percentile-based categorisation and multi-dimensional state classification in subsequent analysis.

DESCRIPIVE STATISTICS GSVA (2023-24)	
MEAN	5527907.901
MEDIAN	3172405
Std. Deviation	5635979.821
MAXIMUM	17628320.7
Minimum	63337.74164
Q1	554534
Q3	9626058.689
IQR	9071524.689

Table 3.1: Descriptive Statistics of State-wise Retail GSVA (2023–24)

DESCRIPIVE STATISTICS GSVA Growth Column	
MEAN	124.267032
MEDIAN	100.5604908
Std. Deviation	93.15083961
MAXIMUM	458.4129814
Minimum	-0.042123782
Q1	87.74249106
Q3	144.4614903
IQR	56.71899924

Table 3.2: Descriptive Statistics of Retail GSVA Growth (%) Across States

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3.5 Descriptive Statistics: UPI Transactions

Financial YEAR	ANNUAL TRANSACTIONS (in Mn.)	ANNUAL VALUE (Rs. in Cr.)
2016-17	**17.86**	**6952.14**
2017-18	**915.22**	**109831.8**
2018-19	**5391.52**	**876970.72**
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2024-25	**185866.02**	**26056954.65**

ABSOLUTE INCREASE	*185848.16*	*26050002.51*
MULTIPLE	*10406.83203*	*3748.048033*
CAGR (Compund Annual Growth Rate) %	*179.491579*	*149.5108645*
CAGR = [(Ending Value / Beginning Value)^(1 / Number of Years)] – 1		

Table 5: National-level summary (financial year table)

This table summarizes national-level UPI activity using annual aggregates of transaction volume and transaction value. Additional indicators such as absolute increase, growth multiple, and compound annual growth rate (CAGR) were calculated to capture the overall magnitude and pace of expansion in digital payment usage. This provides a numerical overview of the growth of UPI before examining any state-level differences..

DESCRIPIVE STATISTICS UPI VALUE(2023-24) [Unclassified not included]	
MEAN	457993.3753
MEDIAN	182326.635
Std. Deviation	648364.5549
MAXIMUM	3168886.74
Minimum	360.02
Q1	15809.5375
Q3	705531.5325
IQR	689721.995

Table 6: State-wise descriptive statistics

Descriptive statistics were also computed for state-wise UPI transaction value for FY 2023–24. Measures of central tendency and dispersion were used to understand how UPI transaction value is distributed across states and the extent of concentration among higher-performing states.

3.6 Summary Insights from Descriptive Analysis

The descriptive statistics indicate wide variation in both retail sector size and digital payment activity across states. Retail GSVA values show high dispersion, reflecting strong concentration of retail output among a limited number of states. Growth percentages also vary considerably, indicating uneven expansion of retail markets.

Similarly, UPI transaction values are highly skewed, with a small group of states accounting for a substantial share of total transaction value. These numerical patterns justify the need for a multi-dimensional analytical framework combining retail size, retail growth, and UPI transaction intensity, which is applied in the subsequent analysis.

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4. Detailed Explanation of Analysis Process / Method

This section outlines the analytical workflow followed to examine the relationship between UPI adoption and Retail GSVA performance across Indian states. The analysis was carried out entirely using spreadsheet-based tools, with each step designed to progressively transform raw data into interpretable business insights. The approach follows a descriptive and classification-based framework rather than a causal econometric model, in line with data availability and the study objectives.

4.1 Data Organisation and Sheet Structure

The analysis was conducted using multiple worksheets, each serving a distinct analytical purpose:

- **Retail GSVA – Raw Data**: Containing state-wise annual Retail GSVA figures across financial years.
- **Retail GSVA – Pivot Summary**: A pivot table consolidating GSVA values by state and year for ease of comparison.
- **Retail GSVA – Working Sheet**: A derived sheet used for growth calculations, size categorisation, and state classification.
- **UPI National Monthly and Annual Sheets**: Capturing monthly UPI transaction volume and value, and their annual aggregates.
- **State-wise UPI Sheet**: Containing UPI transaction value by state for FY 2023–24.

This separation ensured clarity between raw data, intermediate calculations, and final analytical outputs.

4.2 Measurement of Retail GSVA Growth

Retail GSVA growth was measured over the long-term period from 2011–12 to 2023–24 to capture structural trends rather than short-term fluctuations. Two key indicators were computed for each state:

- **Absolute Increase in Retail GSVA**
  Absolute Increase = GSVA (2023–24) − GSVA (2011–12)

- **Retail GSVA Growth Percentage**
  Growth % = [(GSVA (2023–24) − GSVA (2011–12)) / GSVA (2011–12)] × 100

These metrics allowed comparison of both scale-based growth and relative growth performance across states.

Retail_GSVA (Rs. in lakh)	Column Labels						
States	**2011-12**	**2023-24**	**Grand Total (All Years)**	**Absolute Increase (2023-24 value subtracted by 2011-12 value)**	**Growth % [(absolute increase/2011-12 value)*100]**	**Growth Category**	**Size Category**
Maharashtra	10560912.41	17628320.7	198949667.9	7067408.29	66.92043276	Low	High Size

Table 7 : Sample GSVA Growth and Absolute Increase Computation

4.3 Classification of States by Growth and Size

To avoid arbitrary thresholds, percentile-based classification was applied:

- **Growth Category**:
  - High Growth: Above the 75th percentile of growth percentage
  - Medium Growth: Between the 25th and 75th percentiles
  - Low Growth: Below the 25th percentile
- **Size Category**:
  - Based on Retail GSVA levels in 2023–24
  - States were categorised into High, Medium, and Low Size groups using percentile cut-offs

This approach ensured objective grouping based on the empirical distribution of the data rather than subjective benchmarks.

4.4 Computation of UPI Transaction Intensity

UPI adoption at the state level was represented using transaction value share due to the non-availability of consistent historical transaction volume data at the state level.

- **UPI Share (%)** = (State UPI Transaction Value / Total National UPI Transaction Value) × 100

State-wise UPI transaction values for FY 2023–24 were used to compute each state's contribution to national UPI activity.

4.5 Integrated State Classification Framework

The final analytical step combined three dimensions:

1. Retail GSVA Size
2. Retail GSVA Growth Category
3. UPI Transaction Share

Logical conditions were applied to classify states into meaningful strategic segments such as:

- Digitally Mature Markets
- Core Growth Engines
- Emerging Growth Markets
- Low Adoption & Low Maturity States

State / Union Territory	UPI Value (2023-24)	UPI Share % (state UPI value/Total National Value)	Retail GSVA Growth Category	Retail GSVA Size Category	Classification Type
MAHARASHTRA	3168886.74	19.23635672	Low	High Size	Digitally Mature Market

Table 8: Sample Integrated State Classification Structure

This multi-dimensional framework enabled identification of states with differing combinations of economic scale, growth momentum, and digital payment intensity.

4.6 Rationale for Analytical Approach

Given data constraints and the exploratory nature of the study, the methodology emphasizes descriptive analysis and classification rather than causal inference. The objective was to

understand patterns, associations, and relative positioning of states, providing actionable insights for business and policy decision-making. This framework supports comparative assessment of states rather than causal measurement, aligning with the exploratory objective of the study.

5. Results and Findings

This section presents the empirical outcomes obtained from the analytical process described in Section 4. The results are reported in a descriptive and structured manner, focusing strictly on observable trends, distributions, and state-wise outcomes emerging from the data. No interpretation or policy inference is included in this section.

5.1 Trends in Retail GSVA Across States

Retail GSVA data covering the period 2011–12 to 2023–24 indicates a sustained upward trajectory in retail sector output across most Indian states. Over the study period, retail GSVA shows steady long-term expansion, with a temporary moderation during the COVID-19 period, followed by recovery in subsequent financial years.

Considerable inter-state variation is observed in the absolute size of Retail GSVA. Economically larger states such as Maharashtra, Tamil Nadu, Karnataka, Uttar Pradesh, and Gujarat consistently record the highest levels of retail output. In contrast, smaller and less industrialized states exhibit substantially lower GSVA values.

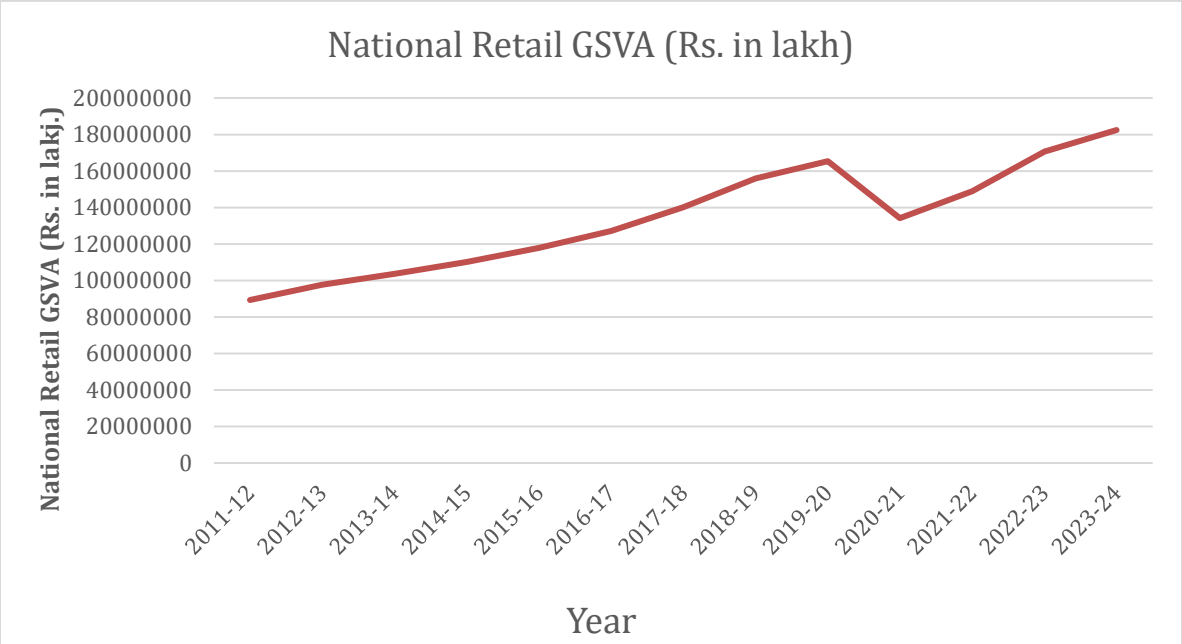

Figure 5.1: Trend in National Retail GSVA

Figure 5.2: State-wise Distribution of Retail GSVA

(Refer to Figure 3: Trend in National Retail GSVA, and Figure 4: State-wise Distribution of Retail GSVA – Top 10 States)

5.2 Retail GSVA Growth Performance

When examined in relative terms, Retail GSVA growth rates vary widely across states. Several medium-sized states demonstrate higher percentage growth compared to larger states, indicating faster expansion from smaller initial bases.

Percentile-based categorisation groups states into high, medium, and low growth categories. States in the high-growth group exhibit strong long-term retail expansion over the study period, while those in the low-growth group show comparatively modest increases.

Figure 5.3: State-wise Retail GSVA Growth Percentage

5.3 National Trends in UPI Transactions

National-level UPI data shows rapid and sustained expansion in both transaction volume and transaction value since its introduction. Monthly trends indicate consistent upward movement across financial years, with moderate seasonal variation.

Annual aggregation confirms strong year-on-year growth in UPI activity. Transaction volumes and values both increase sharply post-2019, with particularly strong acceleration observed after 2020.

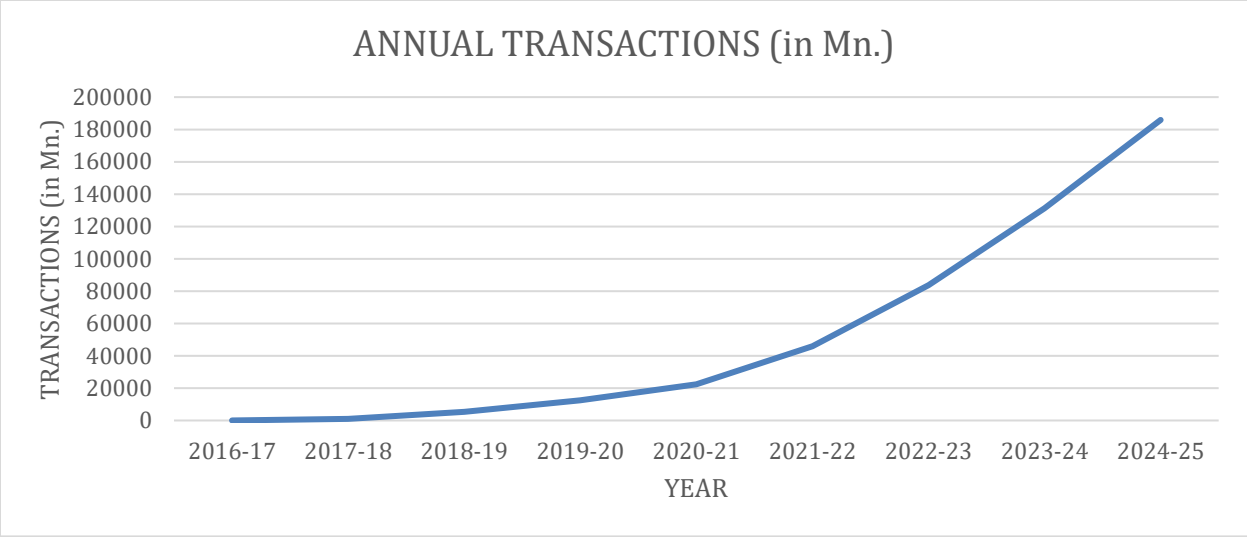

Figure 5.4 : Annual UPI Transactions (Volume)

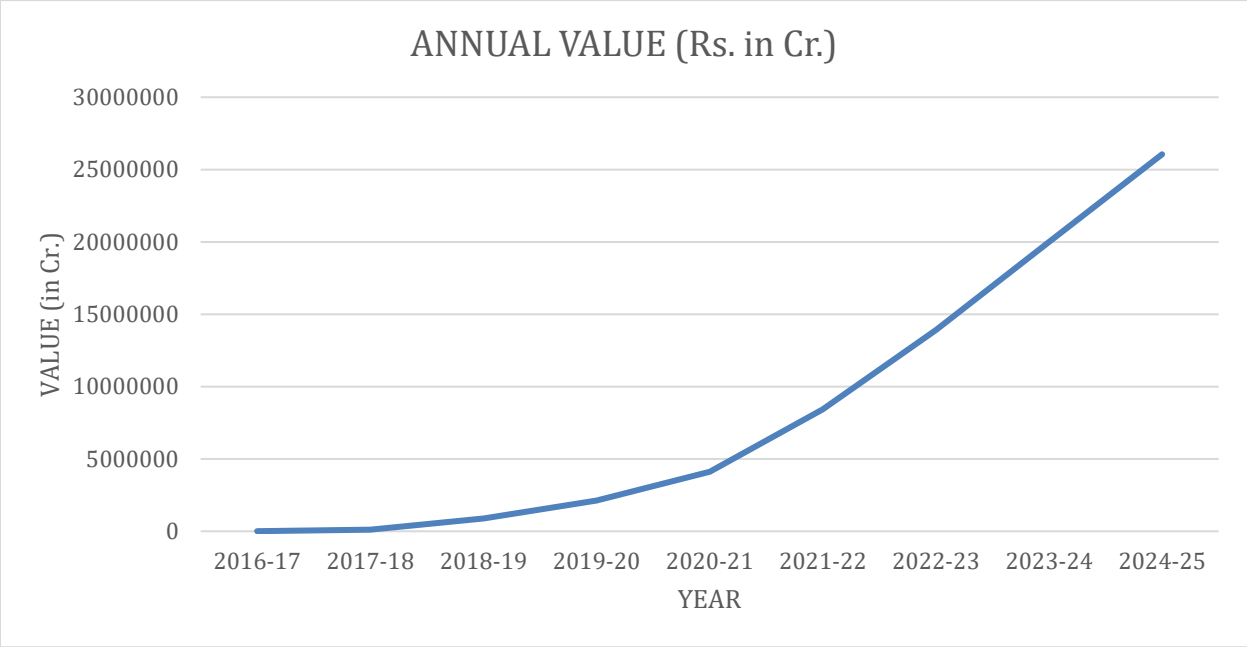

Figure 5.5 : Annual UPI Transactions (Value)

5.4 State-wise Distribution of UPI Transaction Value

State-wise UPI transaction value data for FY 2023–24 displays high concentration. A limited number of states account for a disproportionately large share of national UPI transaction value.

These high-contributing states largely correspond to economically larger states. At the lower end, several states and Union Territories contribute relatively small proportions of total national UPI transaction value.

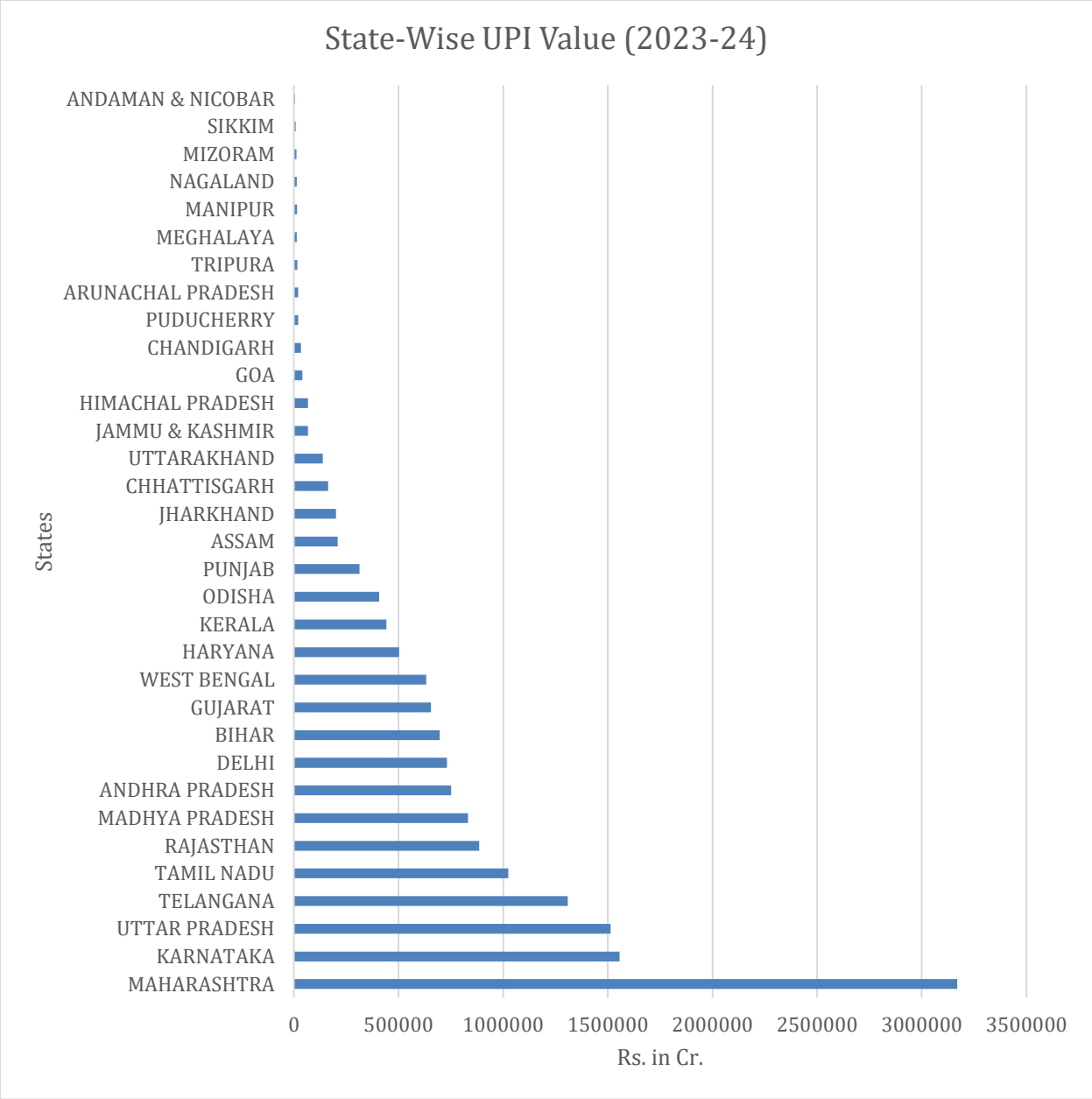

Figure 5.6 : State-wise Distribution of UPI Transaction Value

5.5 Relationship Between UPI Share and Retail GSVA Growth

Comparison of state-wise UPI transaction value share with Retail GSVA growth categories shows the absence of a uniform association pattern. States with high UPI transaction shares are distributed across high, medium, and low retail growth categories.

Similarly, several states with relatively low UPI transaction shares fall within higher retail growth groups. The data therefore reflects variability in growth outcomes across different levels of UPI transaction concentration.

This relationship is examined through cross-comparison of UPI share rankings and Retail GSVA growth classifications.

5.6 State Classification Outcomes

Using the integrated classification framework combining Retail GSVA size, Retail GSVA growth category, and UPI transaction share, all states were grouped into four strategic segments:

• Digitally Mature Markets
• Core Growth Engines
• Emerging Growth Markets
• Low Adoption & Low Maturity States

The classification reveals clear segmentation among states based on differences in economic scale, retail growth momentum, and digital transaction intensity.

Table 9 presents the complete state-wise classification output.

Sr. No.	**State / Union Territory**	**UPI Value (2023-24)**	**UPI Share % (state UPI value/Total National Value)**	**Retail GSVA Growth Category**	**Retail GSVA Size Category**	**Classification Type**
1	MAHARASHTRA	3168886.74	19.23635672	Low	High Size	Digitally Mature Market
2	KARNATAKA	1556774.37	9.45021693	Medium	High Size	Digitally Mature Market
3	UTTAR PRADESH	1513523.11	9.18766521	Medium	High Size	Digitally Mature Market
4	TELANGANA	1307898.79	7.939446799	High	High Size	Core Growth Engine
5	TAMIL NADU	1024543.28	6.219370281	Medium	High Size	Digitally Mature Market
6	RAJASTHAN	885122.88	5.373035031	Medium	High Size	Digitally Mature

						Market
7	MADHYA PRADESH	832228.26	5.051944421	Medium	Medium Size	Digitally Mature Market
8	ANDHRA PRADESH	751565.26	4.562289103	Medium	Medium Size	Digitally Mature Market
9	DELHI	730871.7	4.436671265	Medium	Medium Size	Digitally Mature Market
10	BIHAR	697084.81	4.231571897	Low	Medium Size	Digitally Mature Market
11	GUJARAT	655633.79	3.979948322	High	High Size	Core Growth Engine
12	WEST BENGAL	632247.77	3.83798622	Medium	High Size	Digitally Mature Market
13	HARYANA	502322.19	3.049288167	Medium	Medium Size	Digitally Mature Market
14	KERALA	442393.59	2.685498602	Medium	High Size	Digitally Mature Market
15	ODISHA	407989.27	2.476651197	Medium	Medium Size	Digitally Mature Market
16	PUNJAB	314021.34	1.906229856	Low	Medium Size	Digitally Mature Market
17	ASSAM	208963.05	1.268485781	Low	Medium Size	Digitally Mature Market
18	JHARKHAND	201375.52	1.222426567	High	Medium Size	Emerging Growth Market
19	CHHATTISGARH	163277.75	0.991158505	Medium	Medium Size	Low Adoption & Low Maturity
20	UTTARAKHAND	138981.98	0.843673872	High	Medium Size	Emerging Growth Market
21	JAMMU & KASHMIR	67712.27	0.411039424	Low	Medium Size	Low Adoption & Low Maturity
2	HIMACHAL	67848.2	0.411864814	Medium	Medium	Low Adoption

2	PRADESH	4			Size	& Low Maturity
23	GOA	40899.9	0.248278064	Medium	Low Size	Low Adoption & Low Maturity
24	CHANDIGARH	34537.52	0.20965598	Low	Medium Size	Low Adoption & Low Maturity
25	PUDUCHERRY	20812.17	0.126337846	Low	Low Size	Low Adoption & Low Maturity
26	ARUNACHAL PRADESH	20423.7	0.123979685	Low	Low Size	Low Adoption & Low Maturity
27	TRIPURA	15898.01	0.096507012	High	Low Size	Emerging Growth Market
28	MEGHALAYA	14470.89	0.087843847	High	Medium Size	Emerging Growth Market
29	MANIPUR	15544.12	0.094358764	Medium	Low Size	Low Adoption & Low Maturity
30	NAGALAND	13274.34	0.08058033	Low	Low Size	Low Adoption & Low Maturity
31	MIZORAM	12088.38	0.073381098	High	Low Size	Emerging Growth Market
32	SIKKIM	8903.59	0.054048203	High	Low Size	Emerging Growth Market
33	ANDAMAN & NICOBAR	5306.12	0.032210182	High	Low Size	Emerging Growth Market

Table 9: State Classification Based on Retail GSVA Size, Retail Growth, and UPI Share

This classification forms the empirical basis for interpretation and recommendations in the following section.

6. Interpretation and Recommendations

This section synthesises the key findings of the study in relation to the stated problem statements and outlines focused, data-aligned recommendations. The discussion emphasises direct linkage between the empirical results and their implications for retail sector development and digital payment strategy.

6.1 Interpretation in Relation to the Problem Statements

With respect to the first problem statement, which aimed to assess the impact of state-wise UPI transaction activity on Retail GSVA growth, the results indicate that no consistent or uniform relationship exists across states. Retail GSVA growth was evident prior to the introduction of UPI and continued thereafter, suggesting that underlying structural factors such as income levels, urbanisation, population scale, and consumption demand remain the primary drivers of retail sector expansion. The evidence indicates that UPI adoption is more closely associated with economic scale and transaction formalisation rather than acting as an independent accelerator of retail growth.

Addressing the second problem statement, the integrated classification framework identifies a distinct group of states categorised as Emerging Growth Markets. These states exhibit relatively high retail GSVA growth but comparatively lower UPI transaction shares, indicating untapped potential for deeper digital payment integration. This classification enables systematic identification of such states, allowing targeted strategic focus rather than reliance on isolated indicators.

6.2 Interpretation of State Classification

The state classification highlights distinct economic and digital adoption profiles. Digitally Mature Markets are characterised by large retail sectors and high UPI transaction shares, where digital payments are already embedded in routine retail activity. Core Growth Engines combine substantial market size with strong retail growth, with expansion primarily supported by broader economic momentum. Emerging Growth Markets reflect expanding retail activity alongside relatively lower digital payment intensity, while Low Adoption & Low Maturity states face both economic and digital constraints.

This segmentation demonstrates that states differ not only in retail scale and growth but also in the extent to which digital payments have penetrated retail transactions.

6.3 Recommendations

Based on these findings, the following targeted recommendations are proposed:

• **Prioritise merchant onboarding and digital payment awareness initiatives in Emerging Growth Market states** to complement existing retail expansion and support broader digital integration.

• **Integrate UPI promotion with wider retail modernisation efforts**, including supply-chain digitisation, point-of-sale infrastructure expansion, and logistics support.

• **In Low Adoption & Low Maturity states, focus on foundational economic and infrastructural development alongside phased digital payment adoption**, ensuring readiness before large-scale digital initiatives.

Overall, the results reinforce the view that UPI functions as a complementary enabler within the retail ecosystem, enhancing efficiency and formalisation rather than serving as a standalone driver of retail sector growth.

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7. Limitations and Scope for Future Research

This study is subject to certain limitations that should be considered when interpreting the findings. First, consistent historical state-wise UPI transaction data is not publicly available for the full post-2016 period. Consequently, state-level digital payment intensity is assessed using transaction value share for FY 2023–24, limiting the ability to examine long-term state-wise adoption trends.

Second, UPI transaction value is used as a proxy for digital payment adoption. While this reflects the economic scale of usage, it may be influenced by differences in state size and income levels and may not fully capture adoption depth in terms of transaction frequency, merchant coverage, or user penetration.

Third, Retail GSVA is used as the primary indicator of retail sector performance. Other relevant dimensions such as retail employment, number of retail establishments, and household consumption patterns were not examined due to data constraints.

Future research may extend this analysis by incorporating longer time-series state-wise UPI datasets as they become available, enabling the application of correlation or regression-based methods. Inclusion of additional explanatory variables such as urbanisation, population density, financial inclusion indicators, and infrastructure measures may further strengthen understanding of the relationship between digital payments and retail sector growth.